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Review article

Artificial intelligence across the aging continuum: Mechanistic geroscience, therapeutic innovation, and clinical impact

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ABSTRACT

Aging emerges from nonlinear interactions among primary, antagonistic, and integrative hallmarks that progressively erode tissue resilience. As global demographics shift and chronic disease burden intensifies, extending healthspan with mechanistic precision has become imperative, accelerating the incorporation of artificial intelligence into geroscience. AI leverages multi-omics, spatial biology, imaging, and clinical data to reveal nonlinear structures linking hallmark interactions to tissue vulnerability and organismal decline. These mechanistic insights inform target prioritization, perturbation-based pathway modeling, and rational design of multi-target geroprotectors, including compounds already advancing through clinical trials. Beyond discovery, AI supports synthetic data generation, cross-disease repurposing, and personalized geriatric care through digital phenotyping and predictive analytics. However, these advances hinge on confronting fundamental challenges in data quality, confounding variables, batch effects, and technical artifacts that risk encoding spurious correlations, necessitating hierarchical experimental validation and explainable AI to distinguish causal mechanisms from epiphenomena. Algorithmic bias, digital ageism, privacy vulnerabilities, and infrastructural inequalities further threaten to exacerbate disparities among vulnerable aging populations. This review uniquely traces AI across the complete translational continuum, from hallmark-grounded biomarker discovery to clinical deployment, while positioning validation rigor and ethical infrastructure as core scientific determinants of a transformative AI-geroscience ecosystem.

1. Introduction

With more than 800 million individuals aged 65 and older, the global population is projected to surpass 1.6 billion by 2050. This demographic shift intensifies the burden of chronic non-communicable diseases, frailty, neurodegeneration, and cardiovascular disease, straining healthcare systems and socioeconomic frameworks globally and reinforcing the need for integrated strategies across clinical care, social infrastructure, and research (Fang et al., 2015, 2020, 2025; UN, 2023). Rising longevity has sharpened the dual imperative of extending healthy lifespan and preserving physiological resilience, as reflected in initiatives such as the United Nations Decade of Healthy Aging (2021–2030), which prioritizes functional autonomy and late-life well-being (Bernal et al., 2024; UN, 2023). The persistent gap between lifespan and healthspan has therefore focused attention on aging as the dominant risk factor for chronic disease, multimorbidity, and functional decline across organ systems (Bernal et al., 2024; UN, 2023).

Aging is a complex, multifactorial process driven by the progressive accumulation of molecular and cellular damage that impairs homeostatic resilience and increases susceptibility to chronic disease (Kroemer et al., 2025; López-Otín et al., 2023; Marino et al., 2023). The hallmarks framework organizes this complexity into 12 interconnected processes across primary, antagonistic, and integrative tiers (Kroemer et al., 2025; López-Otín et al., 2013; López-Otín et al., 2023). Primary hallmarks

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initiate the aging cascade through genomic instability, telomere attrition, epigenetic alterations, loss of proteostasis, disabled macroautophagy, and compromised mitophagy, each triggering cellular damage that activates compensatory stress responses. These responses define the antagonistic tier, where deregulated nutrient sensing, mitochondrial dysfunction, and cellular senescence initially buffer damage but become pathogenic when chronically engaged. The cumulative burden of primary insults and maladaptive antagonistic signaling converges on integrative hallmarks, including stem cell exhaustion, altered intercellular communication, chronic inflammation, and dysbiosis, which collectively drive functional decline, multimorbidity, and frailty. The interconnected, nonlinear nature of these hallmarks presents formidable analytical challenges, as perturbations in one domain reverberate across biological scales, producing systemic aging trajectories that elude reductionist approaches and demand integrative computational methods capable of resolving cross-hallmark interactions and predicting intervention responses.

Artificial intelligence (AI) has emerged as a pivotal tool for navigating this complexity. Machine learning (ML), deep learning (DL), and generative AI (GenAI) extract structure from multiomics, imaging, behavioral, and clinical datasets, revealing nonlinear patterns obscured by biological noise, sparsity, and scale (Bernal et al., 2024; Theodorakis et al., 2024; Wilczok, 2025). In geroscience, these approaches are refining biological age estimation, characterizing cellular aging phenotypes, and accelerating biomarker discovery. These capabilities are moving into translational pipelines, where AI nominates targets and shapes intervention strategies around aging biology rather than single diseases in isolation (Theodorakis et al., 2024; Wilczok, 2025; Zhavoronkov et al., 2019). In clinical settings, AI enhances patient stratification, models heterogeneous treatment responses, and predicts trial outcomes, improving the scalability and inclusiveness of clinical trials (Askin et al., 2023; Qian et al., 2025). Beyond formal research environments, AI-enabled wearable sensors, robotics, and ambient monitoring systems support continuous physiological assessment, early detection of functional deterioration, and personalized strategies that help older adults maintain independence (Cantone et al., 2023).

However, the promise of AI in geroscience and geriatric care depends on whether its outputs reflect biology rather than artifacts. Aging datasets are especially vulnerable to confounding, batch effects, population stratification, and technical noise that models may misinterpret as meaningful biological signal (Singh and Benayoun, 2023; Yu et al., 2023). The problem intensifies when high-dimensional molecular data are paired with modest sample sizes, creating substantial overfitting risk in which algorithms capture cohort-specific noise instead of generalizable hallmark biology. Therefore, computational predictions must be validated across cellular systems, model organisms, and independent human cohorts, then interrogated mechanistically to distinguish causal drivers from correlative epiphenomena (Fuellen et al., 2025). Without validation architectures, AI risks generating a proliferation of non-reproducible biomarkers and therapeutic targets that fragment rather than unify the field. Algorithmic bias, privacy vulnerabilities, digital ageism, and opacity further complicate clinical implementation and may exacerbate disparities in older populations while undermining trust in AI-assisted care (Chu et al., 2023; Fahim et al., 2025; Ricciardi Celsi and Zomaya, 2025). Equitable benefits will therefore require robust data quality control, transparent governance, and explainable systems that support accountability and clinical trust.

The central question is not whether AI can be applied to aging research, but whether it can clarify aging biology in ways that are mechanistically credible and translationally useful. Recent reviews have examined AI applications in clinical and translational science (Shahin and Liu, 2025), healthcare delivery (Fahim et al., 2025) and longevity biotechnology (Lyu et al., 2024). The present work takes a different approach by organizing computational advances within the mechanistic framework of primary, antagonistic and integrative hallmarks. This hallmark-centric approach highlights how AI resolves hallmark-specific biological complexities that resist traditional analytical methods. We then trace these insights across the translational continuum, from biomarker discovery and target identification through generative drug design, repurposing, adaptive clinical trials, and precision geriatric care. Critically, we position validation rigor and ethical infrastructure, including data quality control, experimental reproducibility, algorithmic accountability, and equitable access, as scientific determinants of whether AI-geroscience yields reliable and translatable advances. This system-level framing provides a coherent roadmap for an emerging AI-geroscience ecosystem and outlines the conceptual and technical foundations needed to transform aging from a descriptive phenomenon into a modifiable biological process amenable to targeted intervention.

2. AI in biomarker discovery

Biomarker identification is central to drug development and is especially indispensable in geroscience, where biomarkers quantify biological age, frailty, mortality risk, and age-related conditions (Wang and Ji, 2025). Molecular biomarkers are informative because they provide direct readouts of processes embedded within the primary, antagonistic, and integrative hallmarks of aging. (Kroemer et al., 2025; López-Otín et al., 2023). The interconnectedness of these pathways complicates biomarker development that accurately captures the nonlinear, systemic process of aging. This mechanistic complexity is precisely where AI becomes useful. By integrating transcriptomic, proteomic, metabolomic, and other multi-omics datasets, AI extracts high-dimensional mechanistic signals that sharpen pathway-level insight and link molecular signatures with functional phenotypes and longitudinal trajectories. These approaches facilitate the identification of predictive, mechanism-based biomarkers that stratify heterogeneity in aging and guide targeted interventions. This refinement moves biomarker discovery beyond descriptive association toward more precise risk assessment, resilience measurement and therapeutic stratification across the aging spectrum (Pun et al., 2023; Wilczok, 2025).

2.1. AI-enabled biomarker discovery across primary hallmarks

AI is resolving the primary hallmarks of aging with a precision that static molecular assays rarely achieve, converting early molecular lesions into clinically informative patterns across sequence, tissue, and systems scales.

Genomic instability encompasses point mutations, insertions, deletions, microsatellite instability, and chromosomal missegregation that increase structural damage in nuclear and mitochondrial DNA, driving oncogenesis and functional decline. At the detection layer, ML models, including the DNA repair classification signature (DARC Sign) and RF classifiers, trained on multiomics data, detect subtle DNA repair deficiencies (Ritch et al., 2023) and oncogenic mutational signatures, including colibactin mutagenesis in early colorectal carcinogenesis (Rosendahl Huber et al., 2024). A joint single-base substitution (SBS) and insertion-deletion (indel) inference framework sharpens this resolution further, resolving previously confounded recombination deficiency (HRD)-associated signatures into prognostically distinct subtypes, m-SBS3a and m-SBS3b, where m-SBS3b independently predicts improved survival in platinum-treated ovarian cancer cohorts (Ferrer-Torres et al., 2025). DeepHRD extends this into the clinic, identifying 1.8- to 3.1-fold more HRD-positive patients than standard molecular testing and improving stratification of platinum response and overall survival across multiple cohorts (Bergstrom et al., 2024). At the tissue scale, CNNs predict chromosomal instability from nuclear features in whole-slide images (WSIs), correlating high-CIN regions with intra-tumor heterogeneity and adverse clinical outcomes (Hyeon et al., 2024; Xu et al., 2021). MSI-SEER, a deep Gaussian process Bayesian model (Park et al., 2025), and a complementary digital pathology model (Li et al., 2025) extend this spatial logic to microsatellite instability prediction across gastric, colorectal, and gastroesophageal junction

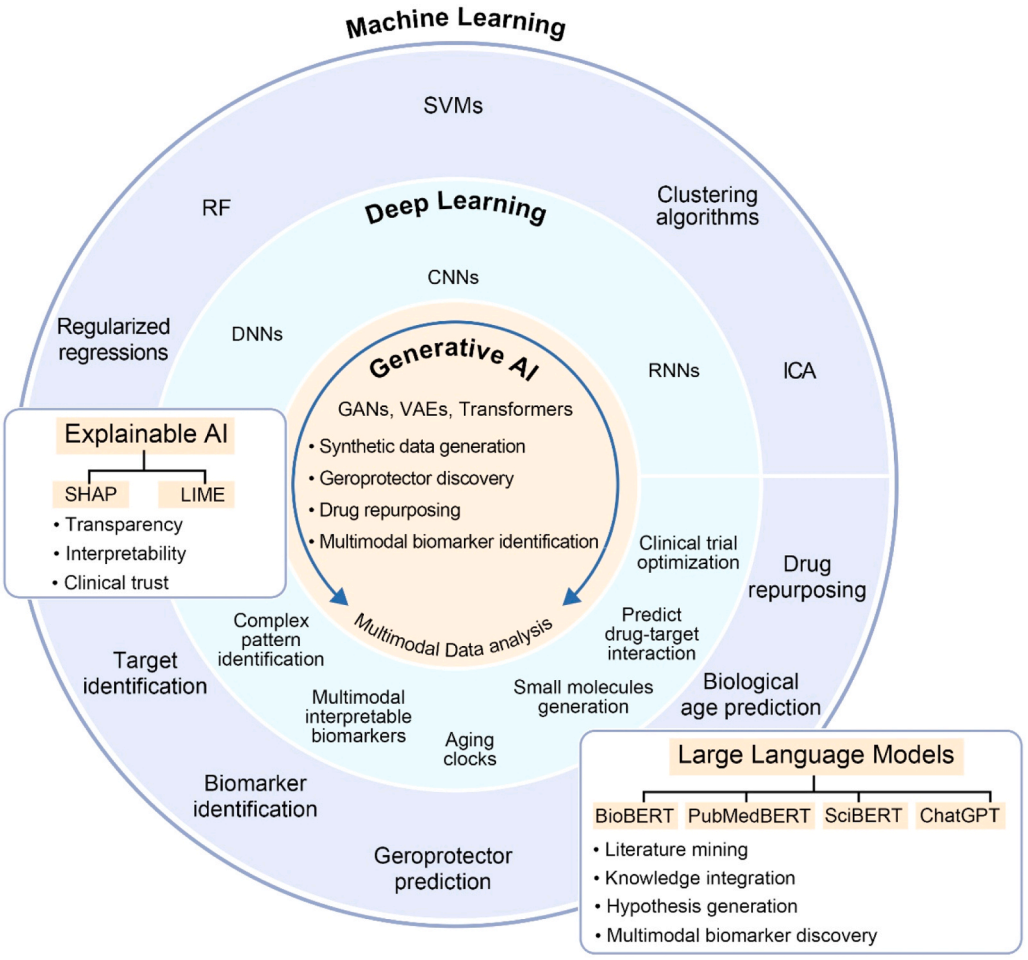


Fig. 1. Hierarchical framework presentation of machine learning (ML), deep learning (DL), and generative AI (GenAI) within aging research. The ML layer is the foundation of the AI architecture, employing supervised algorithms (use labeled data to train models) such as regularized regression, random forests (RF), and support vector machines (SVMs), alongside unsupervised methods like clustering and independent component analysis (ICA) that identify unknown patterns from data. Capable of modeling complex nonlinear aging processes, ML facilitates biological age estimation, target identification, biomarker discovery, health and frailty prediction, and cognitive decline assessment. DL, through neural architectures, including deep neural networks (DNNs), convolutional neural networks (CNNs), and recurrent neural networks (RNNs), captures nonlinear relations across multimodal data such as multiomics, imaging, and longitudinal clinical records, excelling in revealing hidden biological signals, disease classification, and organ visualization. Generative AI models, including generative adversarial networks (GANs), variational autoencoders (VAEs), and transformers, produce realistic synthetic biological, molecular, and clinical data, addressing data scarcity and privacy, thereby advancing biomarker discovery, disease modeling, and drug development through in silico testing and predicting drug-target interactions. Large language models like BERT and ChatGPT facilitate biomedical literature mining, uncovering drug-disease links and biomarkers. Explainable AI methods, such as SHAP and LIME, increase transparency by elucidating feature contributions at both macro and micro levels. Figure assembled and designed using Adobe Photoshop.

cancers, improving immune checkpoint inhibitor response stratification while flagging uncertain slides for expert review.

Telomere attrition compounds this instability, as progressive shortening during replication and cellular stress drive senescence, immune dysregulation, and age-related pathologies, including cardiovascular disease, neurodegeneration, and cancer. Multimodal ML analysis of transcriptomic data uncovers telomere-related gene signatures linking telomere maintenance mechanisms (TMMs) to senescence, immune modulation, inflammation, and tumor heterogeneity (Guo et al., 2024; Li and Liu, 2025). Notably, the ML-derived telomerase signature (MLTS) connects impaired TMM to age-related tissue dysfunction and cancer progression, including increased mutational burden, chromosomal instability and poor survival (Guo et al., 2024). Feature-selection approaches such as RF and SVM further identify regulators, such as MAP7, GRHL2, RARRES2, UCHL1, ESR1, and REV3L, that influence epithelial-mesenchymal transition, angiogenesis, and DNA repair (Zhang et al., 2023). Building on this molecular resolution, TLPath, a DL model, predicts tissue-level telomere attrition directly from H&E-stained biopsies, outperforming chronological age by detecting senescence-like nuclear morphology (Yadav et al., 2026). The AI-enabled TeloSeq platform complements this by resolving full telomere sequence architecture, distinguishing tissue-specific telomere attrition patterns in aging and predicting TMM vulnerabilities for personalized cancer therapy (Schmidt et al., 2024).

Epigenetic modifications, including chromatin remodeling, DNA methylation, histone alterations, and RNA-based mechanisms, couple environmental and physiological factors to gene expression without altering DNA sequence, thereby shaping development and aging. AI platforms such as MethyNet (Levy et al., 2020) and INTERACT (Zhou et al., 2022) leverage methylation data for age prediction, cell-type deconvolution, disease classification, and psychiatric risk assessment. Methylation-based clocks, including AltumAge (De Lima Camillo et al., 2022), DeepMAge (Galkin et al., 2021), and XAI-AGE (Prosz et al., 2024), further estimate biological age across cancer, inflammation, multiple sclerosis, diabetes, and HIV, capturing the molecular drift that accompanies physiological decline (Levy et al., 2020). DunedinPACE adds temporal resolution by quantifying the rate of aging rather than biological state and by uncovering socioeconomic accelerants, including

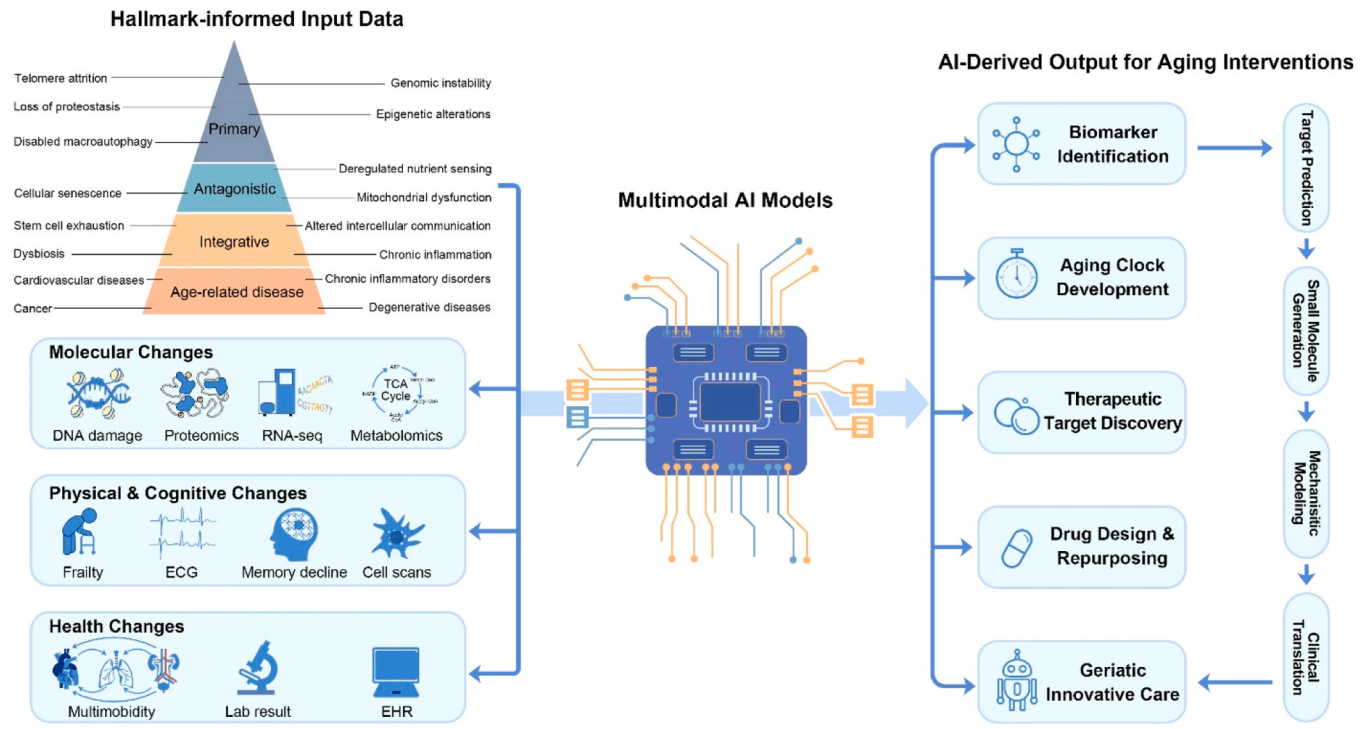


Fig. 2. Schematic illustrating how molecular, physiological, and digital data deriving from the hallmarks of aging converge within multimodal frameworks to generate clinically relevant outputs. Primary aging hallmarks, including genomic instability, epigenetic dysregulation, telomere attrition, proteostasis failure, and autophagy impairment, initiate cellular damage. Their compensatory responses, including deregulated nutrient-sensing pathways, mitochondrial dysfunction, and cellular senescence, afford short-term protection but exacerbate stress over time. The accumulation of these disturbances manifests as integrative hallmarks such as stem cell exhaustion, disrupted intercellular communication, chronic inflammation, and dysbiosis. Multimodal approaches, encompassing multiomics profiling, high-resolution imaging, behavioral phenotyping, and longitudinal clinical data, capture these perturbations across molecular and physiological scales. AI models trained on such datasets enable the identification of biomarkers, the construction of aging clocks, and the discovery of therapeutic targets for precision interventions. By translating complex biological signals into interpretable patterns, this framework elucidates mechanistic links from the molecular underpinnings to clinical applications, thereby advancing the development of targeted, mechanism-based geroscience therapies. Figure assembled and designed using Adobe Photoshop.

poverty and victimization (Belsky et al., 2022). Longitudinal epigenetic clock trajectories predict survival more accurately than single-timepoint measurements, underscoring the value of AI-monitored methylation dynamics over static biomarkers (Kuo et al., 2026). At the chromatin level, the Nucleotide Transformer supports regulatory variant annotation and chromatin state modeling (Dalla-Torre et al., 2025), while supervised ML and DL approaches reveal sequence determinants encoding bivalent histone modifications governing cell fate and disease transitions (Zhao et al., 2026). CrossNN extends this into epigenomic diagnostics by classifying over 170 tumor types from DNA methylation profiles with 99.1% accuracy (Yuan et al., 2025).

Proteostasis decline destabilizes protein synthesis, folding, and degradation, predisposing tissues to misfolding, aggregation, and degeneration (Dalla Bella et al., 2021). Models such as AMYPred-FRL (Charoenkwan et al., 2022) and logistic regression models (Eshari et al., 2023) predict amyloidogenic proteins and aggregation-prone regions based on physicochemical properties. RibbonFold builds on this structural layer by adapting AlphaFold2 to predict the full landscape of amyloid protofilament polymorphs across arbitrary sequences, providing a framework for characterizing aggregation vulnerability and identifying therapeutic targets(Guo et al., 2025). Together, this enables early detection of pathogenic species in amyotrophic lateral sclerosis (ALS), Alzheimer's disease (AD), and Parkinson's disease (PD). AI-QuIC translates this into diagnostic utility by automating misfolded protein seed detection in synucleinopathies through ML-analyzed real-time conversion assays (Howey et al., 2025). At the cellular level, PINNACLE, a geometric DL model trained on a multi-organ single-cell atlas, generates context-aware protein representations across 156 cell types and 24 tissues and improves therapeutic target nomination for rheumatoid arthritis (RA) and IBD (Li et al., 2024). At the systems level, proteomic clocks employing least absolute shrinkage and selection operator (LASSO) regression, gradient boosting decision trees (GBDTs), neural networks, and explainable AI approaches (XAI-SHAP) correlate circulating protein signatures with multimorbidity, mortality, frailty and lifestyle factors across diverse populations (Argentieri et al., 2024; Kuo et al., 2024).

Autophagic disruption closes this hallmark cascade, as age-related decline in protein quality control, organelle turnover, and pathogen elimination drives protein aggregation, inflammation, and metabolic dysfunction. Multimodal DL architectures integrating YOLOv8, U-Net + +, and vision transformers, coupled with XAI class activation mapping (CAM), enable automated high-throughput quantification of autophagic flux in fluorescence imaging datasets, making previously intractable dynamic measurements scalable and interpretable (Presacan et al., 2025). This imaging intelligence is complemented at the molecular level by RF, SVM and LASSO regression that identify autophagy-related diagnostic biomarkers, including EEF2, HSP90AB1, and TNFSF10 in RA, linking impaired autophagy pathways to immune dysregulation that sustains chronic inflammation in aging tissue (Dong et al., 2023).

Across these primary hallmarks, AI transforms dispersed molecular signals into a coherent, multi-layered geroscience framework, where each computational advance builds on the last to convert biological aging complexity into prognostic precision and therapeutic opportunity.

2.2. *AI-enabled biomarker discovery across antagonistic hallmarks*

The dynamic, context-dependent nature of the antagonistic

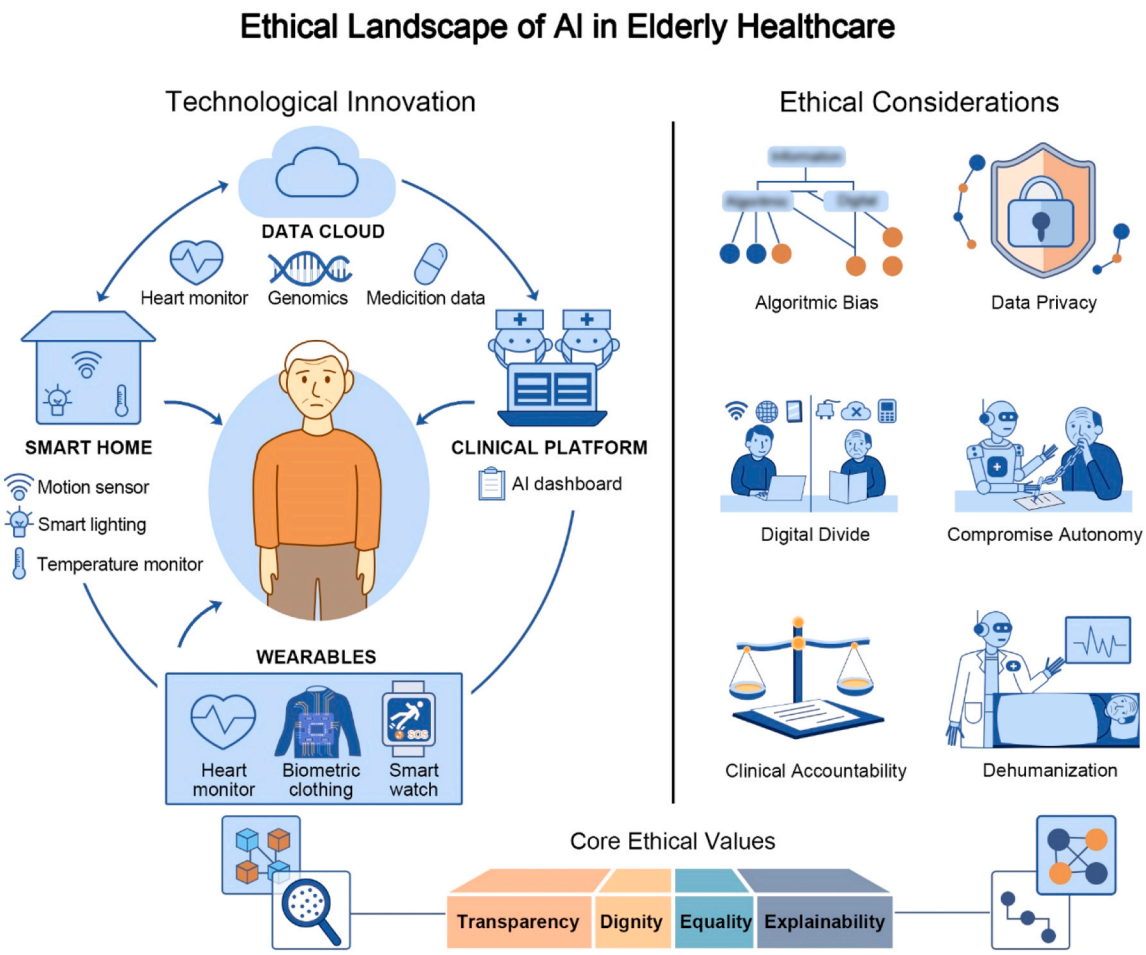


Fig. 3. The ethical landscape of AI-enabled elderly healthcare requires balancing technological advancements with human-centered values. AI-enabled platforms encompass smart-home systems, such as motion sensors, adaptive lighting, and thermal regulation, and wearable biosensors, including ECG-smartwatches, heart rate monitors, and fall detection devices. Cloud-anchored data ecosystems integrate genomic profiles, medication histories, and longitudinal health records into clinical decision support tools that enable real-time, personalized interventions. Ethical considerations, centered on algorithmic bias, privacy, digital exclusion, autonomy, and shared decision-making, are grounded in principles of transparency, dignity, and equity. These principles underpin governance frameworks designed to foster clinician trust and uphold patient agency. As AI becomes integral to geriatric care, the field must ensure that technological innovations substantively enhance care delivery, mitigate disparities, and sustain elder-centered interventions. Aligning AI capabilities with fundamental aging mechanisms and societal factors influencing aging advances the goals of healthy longevity and promotes equitable, biology-informed healthcare tailored to the aging process. Figure assembled and designed using Adobe Photoshop.

hallmarks makes conventional quantification unreliable, while AI provides the analytical frameworks needed to resolve this complexity and transform mitochondrial dysfunction, dysregulated nutrient sensing and cellular senescence into mechanistically interpretable biomarkers.

Mitochondrial dysfunction drives aging through mtDNA mutations, destabilized respiratory complexes, impaired proteostasis, reduced mitophagy, and altered dynamics that collectively impair bioenergetics, elevate reactive oxygen species (ROS), and compromise membrane integrity (Jang et al., 2025). Morphological shifts between elongation and fragmentation render mitochondrial architecture a tractable biomarker and therapeutic target (Jang et al., 2025). DL approaches, including probabilistic interactive models, residual networks (ResNets) and convolutional block attention modules (CBAM), automate mitochondrial segmentation, functional prediction and pathology detection across large imaging datasets (Ding et al., 2025; Jang et al., 2025). Mitochondrial DL (MoDL), trained on super-resolution images linked to biochemical data, extends this capability by predicting membrane potential, ATP production, ROS release, mitophagy, and respiration capacity across diverse cell types (Ding et al., 2025). Moreover, a single-organelle AI strategy integrating fluorescent probes with an RF morphotype classifier distinguishes normoxic from hypoxic mitochondrial states across more than 10,000 organelles, identifying viscosity as a primary functional contributor under stress (Ding et al., 2025). This imaging intelligence connects to therapeutic discovery through an AI-assisted fluorescence microscopy system (AI-FM) that identified Y040–7904 as a mitophagy enhancer attenuating amyloid-β pathology in AD models (Wang et al., 2025). Beyond imaging, ML screens of the mitome siRNA library identify PGS1 as a regulator that restores mitochondrial balance and partially rescues respiration (Cretin et al., 2021). At the diagnostic scale, XGBoost models improve mitochondrial disease prioritization from clinical phenotypes (Kuo et al., 2025); a glmBoost-LDA model achieves a C-index of 0.947 for breast cancer diagnosis from mitochondrial gene signature, with ACADS and AIFM2 as actionable oncological targets (Tong et al., 2025). The MAGIC-H score extends this diagnostic reach to metabolic disease, integrating mtDNA D-loop and ccf-COXIII biomarkers with genetic and anthropometric data to stratify hepatocellular carcinoma (HCC) risk independent of cirrhosis status (Longo et al., 2025). Moreover, multimodal DL frameworks integrating imaging, genomics, and clinical data through attention-based fusion and domain adaptation further refine diagnostics of mitochondrial dysfunction in hematologic cancers (Zhu et al., 2025). Together, these advances establish mitochondria as a

multi-scale biomarker platform where AI converts organelle-level dysfunction into actionable clinical targets.

Dysregulated nutrient sensing across the insulin/IGF signaling (IIS), mTOR, sirtuins, and AMPK axes translates this accumulated damage into gene-expression programs, driving metabolic inflexibility, tissue degeneration, and age-related disease (Petrick and Shomron, 2022). AI interrogates these pathways from food intake to molecular mechanisms, with CNNs (Zeevi et al., 2015) and YOLO architecture (Whig et al., 2025) enabling precise food recognition and individualized post-prandial glucose response prediction. AlphaFold2 (Jumper et al., 2021; Sato et al., 2023) deepens mechanistic insight by resolving conformational states of IIS and mTOR components to reveal pharmacologically relevant regulatory motifs. Multi-omics tools including WiPP (Borgsmüller et al., 2019), Peakonly (Melnikov et al., 2020), ChromAlignNet (Li and Wang, 2019), CANOPUS (Dührkop et al., 2021), DeepMASS (Ji et al., 2019), NEIMS (Wei et al., 2019), MetDNA (Shen et al., 2019) and NormAE (Rong et al., 2020) enhance metabolite annotation, pathway reconstruction and batch correction, thus enabling biomarker identification and accurate mapping of nutrient-sensing networks. Moreover, graph neural networks (GNNs) (Yuan and Nault, 2025) and ensemble frameworks such as TIGER (Han et al., 2022), integrate multi-omics data to elucidate inter-tissue metabolic communication (Chen et al., 2025), aging clocks (Liang et al., 2020) and disease mechanisms in AD, depression, and fatty liver disease, highlighting therapeutic targets including IDO1, LDH, and SGLT2 (Hui et al., 2025; Liang et al., 2020; Takahashi et al., 2020; Varma et al., 2018). XGBoost regression applied to plasma metabolomic and proteomic profiles across 278 mice treated with five anti-aging interventions predicts lifespan extension across intervention classes, with triglycerides carrying longer fatty acid chains emerging as the most influential features (Badenoch et al., 2025). This establishes plasma metabolite patterns as a promising screening tool for anti-aging drug prioritization. The MLP and AdaBoost models incorporating the triglyceride-glucose index as an insulin resistance surrogate predict cardiovascular events in aging populations, directly quantifying how nutrient-sensing dysregulation translates into clinical risk (Alimi et al., 2026).

As metabolic and mitochondrial stress accumulate, cellular senescence emerges as their downstream consequence, characterized by stable cell cycle arrest, macromolecular damage, metabolic reprogramming and a pro-inflammatory senescence-associated secretory phenotype (SASP) (Wang et al., 2024; Zheng et al., 2024). While transient senescence supports wound healing and tumor suppression, chronic accumulation drives inflammation, fibrosis, and degeneration (Wang et al., 2024; Zheng et al., 2024). Traditional markers, including SA-β-gal staining and p16/p21 expression, lack the specificity and throughput to capture this heterogeneity. Tools like SenCID (Tao et al., 2024), SenPred (Hughes et al., 2025), and SenePy (Sanborn et al., 2025), trained on transcriptomics data, classify senescent cells and delineate context-dependent heterogeneity across tissues and inducer types, with SenPred achieving over 99% accuracy in scRNA-seq data from human dermal fibroblasts. DeepScence extends this to spatial resolution, using deep neural networks (DNNs) trained on the CoreScence gene set to identify senescent cells across single-cell and spatial transcriptomics data (Qu et al., 2025). The transcriptomic human universal senescence index (hUSI) complements these tools by quantifying senescence burden across tissues for population-level biomarker identification (Wang et al., 2025). At the image scale, Deep-SeSMo (Kusumoto et al., 2021) and R-CNN (He et al., 2024) detect subtle morphological senescence signatures label-free in microscopy images and real-time senolytics screening. DeepSAHF integrates YOLO-based nucleus segmentation with chromatin pattern classification to enable scalable detection of senescence-associated heterochromatin foci from fluorescence microscopy (Chen et al., 2025). GNNs integrate gene expression (Kim et al., 2025) and topological data (Zeng et al., 2024) to identify regulatory hubs modulating SASP and paracrine signaling. Generative models such as generative adversarial networks (GANs) and autoencoders simulate senescent states in silico (Duran et al., 2024; Duran et al., 2024), thus expanding training datasets and accelerating biomarker discovery. Multimodal transformers trained on transcriptomics, imaging (Fang and Xu, 2025), and methylation (Sidorenko et al., 2024) data uncover tissue- and age-specific senescent phenotypes, while integration of single-cell imaging with mass spectrometry and ML links the metabolomic signature to cellular identity (Kamat et al., 2025; Wang et al., 2025). Together, these AI approaches shift senescent research from static, low-specificity markers to dynamic, systemic-level characterization, thereby revealing regulatory circuitry and senescence trajectories that inform targeted interventions.

Across all three antagonistic hallmarks, AI transforms context-dependent biological responses into quantifiable, spatially resolved, and clinically actionable signatures, completing an arc from organelle-level dysfunction to systemic disease risk stratification.

2.3. AI-enabled biomarker discovery across integrative hallmarks

Damage initiated at primary and antagonistic hallmarks eventually converges at the system level, where stem cell exhaustion, disrupted intercellular communication, dysbiosis, and chronic inflammation emerge as integrative hallmarks that drive multimorbidity and loss of resilience in late life (Kroemer et al., 2025; López-Otín et al., 2023). Their interdependence across tissues makes them difficult to quantify with conventional tools, positioning AI as a vital tool for unraveling underlying mechanisms and predicting how systemic interactions shape aging trajectories.

Tissue regeneration depends on stem cell abundance, functional competence and injury-induced plasticity, which are progressively undermined by cumulative DNA damage, epigenetic alterations, metabolic stress, niche deterioration, and chronic inflammation (Rando et al., 2025). Cellular reprogramming can partially reverse these defects by restoring telomere length, resetting methylation patterns and rejuvenating transcriptomic shifts while preserving cell identity (Browder et al., 2022; Sahu et al., 2024). An OpenAI and Retro Biosciences collaboration advances this further with an AI model that redesigns Yamanaka factor cocktails to reset the biological age by approximately 30 years while maintaining lineage, shifting AI from an analytical tool to an active designer of cellular rejuvenation. CNNs distinguish pluripotent and differentiated states (Kusumoto et al., 2018), classify stem cell types (Liu et al., 2024), and predict intermediate transition states during lineage commitment (Yang et al., 2023), allowing more precise control over reprogramming windows and fate decisions. Furthermore, DeepACT enables reproducible label-free colony selection for regenerative therapy by identifying viable colonies from phase-contrast images through a motion index reflecting central-to-marginal cell movement ratios (Hirose et al., 2021). DL atlases of hematopoietic stem and progenitor cells support aging clocks development and reveal pathways linked to regenerative decline, including cell cycle regulation, cytokine signaling, mitochondrial function, telomere maintenance, and mutational burden (Pham et al., 2025; Zhang et al., 2024). These models estimate donor age and clinical prognosis, uncovering features such as leukemia heterogeneity, youthful myelodysplastic syndrome stem cells and accelerated stem cell aging in sickle cell disease. Tools like GCTx-TFome (Ng et al., 2021), CytoTRACE2 (Kang et al., 2025) and CellRANK2 (Weiler et al., 2024) further reconstruct differentiation trajectories and identify multipotency regulators, rendering stem cell aging quantifiable and strengthening the mechanistic foundation of regenerative medicine.

Aging also erodes intercellular communication by dismantling the mechanosensory, hormonal, and immune systems through which cells interpret their tissue environment. Age-related remodeling of the extracellular matrix (ECM), including collagen, fibronectin, elastin, and glycosaminoglycans, progressively alters the stiffness and traction forces, which cells transduce through adhesion complexes into programs governing morphology, proliferation, and differentiation (Oria et al.,

2025). AI addresses this mechanosensory layer across scales. Feature identification from raw microscopy (FIRM) using RF classifiers identifies fibril architecture that enables systemic characterization of age-related ECM remodeling (Gigliotti et al., 2025), while GANs predict ECM stiffness and cell contractility across tissue contexts (Li et al., 2024). The spatially transformed inferential force map (STIFMap) extends these capabilities by generating spatially resolved maps of stromal mechanical heterogeneity from collagen and nuclear imaging, enabling early detection of pathological changes that precede overt structural failure (Stashko et al., 2023). DNNs further identify adhesion proteins, such as zyxin or paxillin, as traction force predictors (Schmitt et al., 2024), while ML models trained on monolayer stress microscopy data decode how pharmacological perturbations reshape cell-to-cell mechanical signaling, quantitatively linking drug-induced biophysical changes to disease-relevant states (Subramanianbalachandar et al., 2023). Beyond mechanosensory deterioration, aging shifts the balance between pro-aging and rejuvenating circulating signals, impairing tissue repair and regenerative homeostasis, contributing to systemic decline. AI-driven single-cell and spatial-omics frameworks, including CellChat (Jin et al., 2025), DeepCCI (Yang et al., 2023), and SpaCcLink (Liu et al., 2025), delineate ligand-receptor interactions, such as BMP7–BMPR1a/BMPR2, CCL19-CCR7, MIF–CD74/CD44, CXCL12-CXCR4, and VEGFA–VEGFR2, and transcriptional regulators, including FOXO1/3, Trp53, and NFKB1, that coordinate inflammatory and regenerative responses. The scAgeCom atlas, built from 58 single-cell datasets across 23 mouse tissues, transforms these signals into a body-wide map of communication collapse, revealing upregulated inflammatory signaling, suppressed angiogenesis, and dysregulated lipid metabolism as universal aging signatures (Lagger et al., 2023). PathFinder's graph-transformer architecture extends this to disease contexts, identifying APOE4-associated networks in AD to CCR9/CCL25 and CCR6/CXCL16 axes in cirrhosis, implicating autophagy, lipid metabolism, and JAK-STAT signaling as age-sensitive pathways amenable to intervention (Feng et al., 2024).

Age-related dysbiosis adds a microbial dimension to these system-level changes. Reduced microbial diversity, loss of beneficial taxa, and expansion of pathobionts disrupt metabolic and immune regulation, actively amplifying inflammaging, frailty, and susceptibility to neurodegenerative and cardiovascular diseases. Healthy aging consistently associates with microbiomes enriched in anti-inflammatory, antimicrobial and antioxidant metabolites, whereas deleterious metabolites like p-cresol correlate with frailty (Pu et al., 2024; Sato et al., 2021; Wu et al., 2022), underscoring dysbiosis as an active driver rather than a mere biomarker of systemic aging. AI is transforming microbiome research from taxonomic description into mechanistically grounded, predictive models (Zhou and Zhao, 2025). GBDTs and XAI-SHAP analysis identify microbial and metabolic signatures that stratify individuals by cognitive resilience or vulnerability (Kim et al., 2024). DNNs and ensemble ML models further integrate microbial profiles with host gene expression to construct aging clocks mapping biological aging trajectories (Galkin et al., 2020; Gopu et al., 2024; Wang et al., 2024). DysbioFormer, a dysbiosis-aware multiset transformer, detects imbalance as an emergent, community-scale property, moving beyond single-taxon associations to capture system-level microbial imbalance across disease states (Alakilli et al., 2026). ML-driven meta-analysis harmonizing independent PD cohorts identified robust dysbiotic signatures, including *Akkermansia* enrichment and depletion of butyrate-producing *Roseburia* and *Faecalibacterium prausnitzii*, establishing causally informative microbial correlates of neurodegeneration at the population scale (Romano et al., 2025). Self-supervised microbial language models extend mechanistic depth further by encoding each taxon contextually within its broader community architecture, revealing short-chain fatty acid and indole signaling pathways through which dysbiosis drives host inflammatory tone and neurodegeneration (Lee et al., 2025). Clinically, Q-net, a generative AI digital twin trained on longitudinal infant microbiome data, predicts neurodevelopmental risk with 76% accuracy and proposes taxon-specific restoration strategies that attenuate deficits in approximately 45% of at-risk individuals. Critically, the model also demonstrates that mismatched interventions worsen outcomes, underscoring the necessity of AI-guided individualization over empirical supplementation (Sizemore et al., 2024).

Inflammaging, a chronic, low-grade inflammatory state characterized by elevated IL−6 and CRP, immune dysregulation, and T cell subset shifts, arises from genomic instability, cellular senescence, mitochondrial dysfunction, and autophagic disruption (Kushner et al., 2025). This self-sustaining inflammatory milieu accelerates cardiovascular disease, neurodegeneration, and metabolic dysfunction. The Canakinumab Anti-Inflammatory Thrombosis Outcomes Study (CANTOS) trial demonstrates that targeting inflammation can reduce cardiovascular events, diabetes incidence, and malignancy, establishing inflammation as a tractable driver of age-related disease rather than a secondary correlate (D'aiello et al., 2024). AI now underpins multiscale quantification and therapeutically targeting of inflammaging across biological scales. The inflammatory aging (iAge) clock, built with a deep neural autoencoder, quantifies chronic inflammatory burden independent of chronological age and identifies CXCL9 as an early biomarker for vascular and immune deterioration (Sayed et al., 2021). The EpInflammAge clock deepens this resolution by integrating DNA methylation with cytokine and chemokine signatures through DANet and SHAP-based interpretability, capturing nonlinear aging trajectories and generating individual-level cytokine predictions from methylation inputs overlooked by traditional clocks (Kalyakulina et al., 2025). At the cellular level, SVR trained on mass cytometry data across 30 protein markers and six major immune subsets predicts immunological age and has been validated in a murine model of metabolic dysfunction-associated immune senescence, providing a translational framework for quantifying immune aging trajectories (Sim et al., 2025). At the tissue scale, StainAI maps microglial inflammatory states across whole brain sections from standard histology at speeds orders of magnitude faster than manual scoring and reveals region-specific neuroinflammatory vulnerabilities after cardiac arrest that would otherwise remain undetected (Hsu et al., 2025). Together, these AI-enabled inflammatory clocks and profiling tools translate immune architecture into mechanistically grounded metrics across molecular, cellular, and tissue scales, positioning chronic inflammation as a modifiable hallmark and strategic target for reducing multimorbidity and frailty.

3. AI-driven drug discovery and drug repurposing

AI has reshaped aging therapeutics by enabling precise target identification, accelerating molecule discovery, and optimizing clinical trial design. Its capacity to interrogate high-dimensional molecular, physiological, and clinical datasets is shifting drug discovery from hypothesis-driven approaches to adaptive, data-driven strategies that better capture the complexity of aging biology. The application of AI to geroprotector discovery has emerged as a major translational frontier, with mechanism-based approaches targeting conserved longevity pathways now recognized as essential for durable clinical translation (Morsli and Bellantuono, 2021; Moskalev, 2023). Recent AI-driven platforms have demonstrated proof-of-concept across the discovery pipeline, from target identification through lead optimization and clinical translation, by computationally integrating multi-omics data, chemical libraries, and phenotypic readouts to nominate compounds that engage fundamental aging mechanisms.

3.1. Target identification

Target identification involves recognizing biological nodes that can be therapeutically modulated, supporting strategies that modify aging biology rather than targeting individual diseases. Traditional reductionist approaches that interrogate single pathways in isolation have struggled to capture the network architecture of aging. AI integrates

omics, molecular networks, and clinical phenotypes through network medicine, GNNs and large language models (LLMs) to prioritize targets at the intersection of aging circuits and age-related disease (Pun et al., 2026). This shifts the focus from organ-specific targets to dual-purpose nodes that influence both disease expression and core hallmarks of aging.

This dual-purpose paradigm is exemplified by PandaOmics, which analyzed thousands of healthy and tumor transcriptomes to identify age-associated cancer targets, validating that inhibiting KDM1A extends *Caenorhabditis elegans* lifespan while suppressing colorectal and triple-negative breast cancer (Pun et al., 2023). Building on this, PandaOmics and Chemistry−42 identified cyclin-dependent kinase 20 (CDK20) as an HCC target, leading to the discovery of ISM042–2–048 with antiproliferative activity (Ren et al., 2023). PRPF19 and MAPK9 subsequently emerged as dual targets for HCC and cellular senescence, while CNGA3, GLUD1, and SIRT1 were identified as candidates bridging glioblastoma and aging pathways (Olsen et al., 2023). This oncology framework extends into immuno-oncology through MIDAS, a multimodal GNN integrating gene interactions, multi-omic patient profiles, single-cell immune atlases, CRISPR perturbation data, and HLA-peptidomics to prioritize immunotherapy targets (Augustine et al., 2026). MIDAS nominated the OSM-OSMR axis as a candidate immunotherapy target, and its perturbation in TRACERx melanoma patient-derived explants reduced dysfunctional CD8 + T cells, thereby validating an AI-based target in a clinically relevant platform (Augustine et al., 2026).

Beyond oncology, AI-driven target discovery is accelerating development for neurodegenerative diseases, where aging is the dominant risk factor. GuiltyTargets applied network representation learning to a genome-wide protein-protein interaction network annotated with disease-specific gene expression, prioritizing AD targets including CHRFAM7A, GRM1, GRM3, ITPR1, HTR7, and COMT (Ö et al., 2022). In amyotrophic lateral sclerosis (ALS), an ML-augmented genome-wide association study (GWAS) approach identified KANK1 as a novel causal gene and reproduced patient mutations in human neurons, connecting genetic risk to cellular pathology (Zhang et al., 2022). PandaOmics further integrated multi-omics and natural language processing (NLP) across central nervous system tissues and induced motor neuron models, nominating panels of ALS targets that were validated in *C9ALS* Drosophila models (Pun et al., 2022). These studies demonstrate how AI connects genetic epidemiology, network context, and experimental validation across the neurodegeneration–aging axis.

AI-driven target discovery extends into cardiovascular aging, where mechanistic complexity has hampered therapeutic development. DL phenotypic screening in BAG3-deficient iPSC-derived cardiomyocytes identified HDAC6 inhibition as cardioprotective, with TYA−018 reducing sarcomere damage and apoptosis while improving mitochondrial function through upregulation of fatty acid oxidation and oxidative phosphorylation (Yang et al., 2022). These mechanisms underpin age-related cardiac decline, illustrating how AI links organelle-level dysfunction to actionable targets. At the population scale, DL brain age models trained on UK Biobank MRI, combined with GWAS, Mendelian randomization, and colocalization analysis, prioritized seven genetically supported druggable genes, including MAPT, TNFSF12, and GZMB, as therapeutic targets for brain aging (Yi et al., n.d). This framework also identified 13 repurposable clinical-stage drugs with genetic backing, ensuring target selection is grounded in human causal evidence.

AI further reveals that the most effective geroprotectors act polypharmacologically rather than on single targets. GNNs trained on the lifespan-extending compound space show that the most efficacious compounds simultaneously engage multiple biogenic amine G protein coupled receptors (GPCRs), and more than 70% of predicted candidates extend *C. elegans* lifespan (Avchaciov et al., 2025). AURORA scales this insight systemically with a generative DL platform that integrates transcriptomics, metabolomics, microbiome, clinical laboratory tests and facial imaging data from over 581,000 samples. This approach harmonizes batch effects, reconstructs missing modalities, and predicts intervention responses validated in longitudinal cohorts (Chen et al., 2026). Collectively, these advances resolve the multiscale causal architecture of aging into actionable therapeutic nodes, enabling interventions that target root mechanisms rather than isolated downstream symptoms.

3.2. Drug-target interaction

Drug–target interaction (DTI) prediction bridges target identification and gerotherapy design by identifying compounds that modulate aging-related nodes with adequate potency and selectivity. Conventional approaches struggle with sparse interaction matrices and protein cold start, leaving many potentially geroprotective interactions unexplored. The knowledge graph-neural factorization machine (KG-NFM) framework mitigates this by embedding drugs, proteins, diseases, and side effects within a unified knowledge graph and integrating these with a neural factorization machine, enabling prediction of novel DTIs even for poorly characterized proteins (Ye et al., 2021). Recent DL surveys further show a shift from simple similarity models toward GNNs and transformer architectures that learn joint drug–target representations, improving the generalization for complex, multi-hallmark traits such as aging (Chen et al., 2025).

Building on this foundation, a second wave of DTI models enhances interpretability and uncertainty, crucial for de-risking geroscience applications. Interpretable nested GNNs for DTI (iNGNN-DTI) (Sun et al., 2024) and graph-based attention models such as AttentionSiteDTI (Yazdani-Jahromi et al., 2022) exploit protein binding sites and self-attention to identify drug substructures and pockets that drive predicted interactions, linking in silico hits to pathways such as senescence, mitochondrial, or inflammatory pathways rather than relying on opaque scores. Evidential DL DTI (EviDTI) frameworks attach calibrated uncertainty to each prediction, enabling prioritization of candidate senolytics or mitoprotective agents with high confidence, while diverting uncertain predictions to orthogonal validation pipelines (Zhao et al., 2025). These advances transform DTI models from hit rankers into mechanistically informative filters that enrich for multi-target geroprotectors while constraining off-target risk (Chen et al., 2025).

Geroscience-specific pipelines are now leveraging this capability to nominate intervention candidates. DRADTiP integrates network representation learning with DTI prediction to link aging-related genes and approved drugs, highlighting repurposing candidates such as metformin, aspirin, and statins as potential interventions for age-associated disease (Muniyappan et al., 2024). The generative docking platform DiffDock further exemplifies this potential by elucidating how the narrow-spectrum antibiotic enterololin targets the LolCDE lipoprotein transport complex in Crohn's-associated *Escherichia coli*, compressing mechanism-of-action discovery from years to months and validating AI-driven target engagement in vivo (Catacutan et al., 2025). This progress suggests similar approaches could map senolytics, mitoprotective agents, and metabolic modulators onto aging processes for rational gerotherapy development.

3.3. Accelerated geroprotector discovery

Geroprotectors are agents that target fundamental aging mechanisms such as oxidative stress, mitochondrial dysfunction, chronic inflammation, and cellular senescence to extend healthspan rather than merely treat age-associated diseases (Morsli and Bellantuono, 2021; Moskalev, 2023). Unlike traditional drugs, they modulate key biological pathways to delay multiple age-related pathologies. Major classes include metabolic modulators, senolytics, antioxidants, autophagy enhancers, epigenetic regulators, mitochondrial stabilizers, and anti-inflammatory compounds (Morsli and Bellantuono, 2021; Moskalev, 2023; Partridge et al., 2020). Preclinical studies show that these agents enhance

mitochondrial function, clear dysfunctional cells, and activate autophagy pathways (Morsli and Bellantuono, 2021).

Traditional discovery methods, such as lifespan assays and transcriptomic profiling, are resource-intensive and limited in scope. AI partly mitigates these constraints by integrating chemical structures, multi-omics data, and functional annotations to identify lifespan-extending candidates. For example, AgeXtend leverages XAI to evaluate over a billion compounds, resulting in novel geroprotectors validated across biological models (Arora et al., 2025). Similarly, GeroScope employs DNNs to identify compounds that activate youthful gene expression programs, exemplified by PD−98059, myricetin, EGCG, and N-acetyl-L-cysteine, which rejuvenate senescent cells in vitro (Aliper et al., 2016). Advanced AI models have also uncovered mTORC1 inhibitors, such as TKA001, trihydroxysterol and ganoderic acid derivatives that replicate rapamycin's anti-aging effects with fewer adverse outcomes, thereby expanding the pharmacological toolkit for restoring metabolic and nutrient-sensing homeostasis (Posansee et al., 2023; Vidovic et al., 2023; Yu et al., 2025). Furthermore, multimodal ML architectures and DNNs classify AMPK activators with high accuracy from over 1700 compounds, enabling rapid in silico prioritization of modulators (Drewe et al., 2021). GNN-based models such as TxGNN predict drug-disease interactions without prior pairwise training, successfully identifying compounds targeting nuclear instability, cellular senescence, and DNA damage in progeria syndrome (Huang et al., 2024; Muniyappan et al., 2024). These frameworks provide a system-level framework for translating network-based predictions into geroscience interventions.

AI-driven screening has also yielded promising geroprotector candidates from chemical and natural compound libraries. For instance, ML analysis of mitochondrial quality control pathways identified kaempferol and rhapontigenin as mitophagy-inducing compounds that ameliorate AD pathology in preclinical models, reducing Aβ and tau accumulation, neuroinflammation, and cognitive decline (Xie et al., 2022). More recently, α-amyrin was identified as a complementary mitochondrial "guardian" that enhances ULK1-dependent mitophagy via modulation of the DLK–SARM1 axis, restoring mitochondrial homeostasis and mitigating AD-like phenotypes in vivo (Cao et al., 2026).

The convergence of AI and senescence biology is similarly reshaping senolytic discovery. First-generation senolytics such as dasatinib and quercetin demonstrated the feasibility of clearing senescent cells but are constrained by toxicity, off-target effects, and poor bioavailability (Gadecka et al., 2025; Islam et al., 2023; Nambiar et al., 2023). AI-driven screening using CNNs and GNNs trained on high-throughput imaging and chemical libraries has accelerated the discovery and optimization of more selective compounds, including ginkgetin, oleoandrin, periplocin, panaxatriol, and voclosporin, which show improved pharmacokinetics and more precise engagement of senescence pathways (Li et al., 2025). Morphology-based CNN models and Deep-SeSMo screening identify agents such as terreic acid and daidzein that reduce senescent cell burden and suppress SASP activity, while preserving beneficial aspects of senescence (Kusumoto et al., 2021). Moreover, GNNs have also pinpointed compounds such as BRD-K56819078, BRD-K20733377, and BRD-K44839765, which reduce senescent cell burdens in aged murine models with favorable pharmacological profiles and cellular selectivity (Wong et al., 2023).

Together, these advances underscore an emerging paradigm toward mechanistically precise, AI-guided geroprotective therapeutics. By integrating molecular, cellular and system-level data, AI enables the rational design of interventions that target core aging processes, offering a path toward safer, more effective strategies to delay physiological decline and extend healthspan.

3.4. Generative AI-derived drugs

Generative AI (GenAI) architectures, including variational autoencoders (VAEs), GANs, and reinforcement learning systems, enable de novo design of molecules with optimized structural and pharmacological profiles (Pun et al., 2023). Beyond real-world omics and screening data, these models increasingly leverage synthetic datasets, engineered to mimic biological patterns, allowing the simulation of diverse biological states, thus expanding the scope of exploratory research in aging and age-related diseases (Mishra et al., 2025; Shi et al., 2020; Wilczok, 2025). This strategy is particularly valuable in rare geropathologies or conditions with limited patient cohorts, where empirical datasets are scarce (Pun et al., 2023). By encoding preexisting biological knowledge, these datasets support model training for therapeutic discovery, pathway inference and mechanistic hypothesis generation, while also serving as validation layers that strengthen confidence in target prioritization.

Generative models excel at navigating complex molecular networks, identifying synergistic target combinations suited to multifactorial aging phenotypes. For instance, the generative tensorial reinforcement learning (GENTRL) generated potent DDR1 kinase inhibitors in under two months from design to experimental validation, illustrating AI's capacity to accelerate discovery while reducing costs in geroscience research (Zhavoronkov et al., 2019). Furthermore, generative chemical language models (CLMs) extend this paradigm by generating multi-target, drug-like molecules that reproduce key

pharmacophore features of known ligands while incorporating activity data and polypharmacology constraints. These models have produced dual ligands targeting FXR and sEH, FXR and THRβ, and PPARδ and sEH, targeting pathways central to metabolic syndrome, such as hypertension, non-alcoholic fatty liver disease, and insulin resistance (Isigkeit et al., 2024). However, generative models carry inherent risks, including mode collapse, hallucination of chemically implausible structures, and the potential for synthetic training data to introduce systematic biases that inflate in silico performance without corresponding in vitro validity.

3.5. AI-enabled drug repurposing

Aging and chronic disease share intricately connected molecular pathways, rendering dual-target and pathway-centric strategies particularly compelling. AI accelerates drug repurposing by identifying compounds that modulate conserved aging-related pathways and leveraging existing pharmacophores, thereby improving safety, minimizing preclinical validation burden, and establishing scalable, reproducible pipelines. These strategies facilitate rapid translation of mechanistic insights into interventions capable of addressing multisystem decline.

The GuiltyTarget framework integrates disease-specific transcriptomics with genome-wide protein-protein interaction (PPI) networks to prioritize repurposing candidates. Notably, it identified catechol-O-methyltransferase (COMT) as a potential therapeutic AD target, despite COMT's established role in tolcapone-based PD therapy, illustrating cross-disease target reallocation when neurodegenerative mechanisms converge (Ö et al., 2022). ML analysis of transcriptomics and pathway data similarly identified tissue-specific therapeutic targets such as carbonic anhydrase IV (CA4), suggesting that agents like topiramate and chlorothiazide may warrant evaluation in neuromuscular disorders (Mamoshina et al., 2018). DNNs trained on transcriptional drug response data propose reapplication of drugs like otenzepad and pinacidil for central nervous system therapeutics (Aliper et al., 2016). AI frameworks have similarly been applied to drug development for delirium and AD, identifying candidate compounds through integrated target network and transcriptomic analyses (Ai et al., 2025). Moreover, ML-driven propensity score models, combined with inverse treatment weighting applied to healthcare claims, implicated pantoprazole, gabapentin, atorvastatin, fluticasone, and omeprazole as potential AD interventions (Zang et al., 2023). Complementary network-based AI frameworks have also identified lipid regulator gemfibrozil and cardiovascular agent simvastatin as potential interventions for AD (Xu et al., 2022) and PD (Dou et al., 2025), respectively. By delineating

conserved molecular patterns across omics and clinical datasets, AI accelerates the discovery of broad-spectrum therapies targeting aging-related phenotypes and supports multi-indication, mechanism-anchored strategies to extend healthspan. Nevertheless, most AI-repurposed drugs fail at experimental validation, underscoring that network proximity or transcriptomic concordance is a starting point rather than sufficient evidence of therapeutic effect.

3.6. Clinical trial optimization

AI is reshaping clinical trial design by streamlining protocol development, outcome modeling and operational execution. Multimodal integration of genomics, cheminformatics, literature, and preclinical datasets enables predictive models that forecast trial outcomes with up to 79% accuracy, thus improving resource allocation and de-risking aging-focused therapeutic development (Huang et al., 2025). Moreover, NLP tools further translate complex protocols into operational workflows, thereby reducing implementation delays and enhancing trial fidelity (Schopow et al., 2023).

EHR-based models further optimize recruitment by identifying eligible participants, enriching underrepresented cohorts, and refining phenotypic inclusion criteria (Liu et al., 2021). For instance, the Trial Pathfinder framework demonstrated that data-informed modifications can broaden enrollment in non-small cell lung cancer (NSCLC) trials without compromising scientific rigor, serving as a proof of concept for geroscience trials where heterogeneity and comorbidity often limit participation (Liu et al., 2021). Collectively, these innovations position AI as a key catalyst for more efficient, representative, and mechanistically informed clinical studies for age-related therapeutics.

3.7. AI-derived drugs in clinical trials

AI-designed molecules are now entering human trials, marking a pivotal transition for computational prediction to clinical translation. Companies such as Exscientia, Insilico Medicine, Evotec, and Schrödinger have advanced multiple candidates into early-phase studies targeting cancer, neurodegeneration, fibrosis, metabolic syndromes, and ophthalmologic, underscoring the breadth of AI's application to conditions tightly linked to aging biology (Arnold, 2023).

Exscientia's EXS−21549, a selective A2A antagonist, is in Phase 1/2 trials (NCT04727138) for relapsed renal cell carcinoma and NSCLC. Similarly, Insilico Medicine's INS018_055, a TNIK inhibitor for idiopathic pulmonary fibrosis, recently completed Phase 2a trials (NCT05938920), demonstrating anti-fibrotic and anti-inflammatory activity. Additional AI-designed agents, including DSP−0038, a dual serotonin receptor modulator, and EXS4318, a PKCθ inhibitor, are progressing through early-phase trials for AD psychosis and inflammatory conditions, respectively. GSBR−1290 (aleniglipron), an oral GLP−1 receptor agonist, and OPL−0401, a ROCK1/2 inhibitor, are in Phase 2 trials for obesity (NCT06703021) and diabetic retinopathy (NCT05393284). Firsocostat (GS−0976), an acetyl-CoA carboxylase inhibitor, is under clinical investigation (NCT02876796) for NASH.

These trials are beginning to define the pharmacokinetics, safety, and pharmacodynamic profiles of AI-derived therapeutics. Because their targets converge on immune exhaustion, chronic inflammation, metabolic dysregulation, and tissue fibrosis (Table 1), they position AI-guided drug design as a credible translational framework for geroscience. Explicitly embedding geroscience endpoints into such pipelines could accelerate interventions that stabilize tissue homeostasis, preserve immune competence, and mitigate metabolic decline across the lifespan. This potential will only be realized, however, if training datasets reflect the demographic heterogeneity of aging populations rather than reinforcing the enrollment biases that have historically limited geriatric trial generalizability. Collectively, these advances signal a paradigm shift in which AI-driven platforms are no longer purely predictive tools but validated engines of translational insight. By systematically identifying and repurposing compounds that modulate the molecular hallmarks of aging, they demonstrate an efficiency that

Table 1
Representative AI-identified drug candidates and therapeutic targets with documented in vitro, preclinical, or clinical validation for aging and age-related diseases.

Drug / Candidate	AI Method	Target / Pathway	Age-related Indication	Reference
In vitro validated candidates				
PD−98059, Myricetin, EGCG, NAC	DNN (GeroScope)	Youthful gene expression reactivation	Cellular senescence	(Aliper et al., 2016)
ISM042–2–048 (inhibitor)	PandaOmics & Chemistry42	CDK20	HCC, cellular senescence	(Ren et al., 2023)
TYA−018 (inhibitor)	DL	HDAC6	Age-related cardiomyopathy	(Yang et al., 2022)
DDR1 kinase inhibitors	GENTRL	DDR1 kinase	Fibrosis, aging-related inflammation	(Zhavoronkov et al., 2019)
TKA001 (inhibition)	GAN and reinforcement learning	mTORC1	Aging, metabolic dysfunction	(Vidovic et al., 2023)
Preclinical validated candidates				
Kaempferol, Rhapontigenin	Unsupervised ML, network analysis, pharmacophore screening	Mitophagy	Alzheimer's disease	(Xie et al., 2022)
α-Amyrin	AI-driven systems pharmacology	DLK–SARM1–ULK1 axis (mitophagy)	Alzheimer's disease	(Cao et al., 2026)
Y040–7904	AI-FM screen	Mitophagy enhancer	Alzheimer's disease	(Wang et al., 2025)
Voclosporin	SVM and MLP models with MoLFormer embeddings	Senolytic activity	Cellular senescence, aging	(Li et al., 2025)
BRD-K56819078	GNN screening	BCL2 inhibition	Aging, multimorbidity	(Wong et al., 2023)
Clinical-stage candidates				
INS018_055	Generative AI (InSilico Medicine)	TNIK inhibition	Idiopathic pulmonary fibrosis	NCT05938920
EXS−21549	AI-guided design (Exscientia AI)	A2A receptor antagonism	RCC, NSCLC	NCT04727138
Firsocostat (GS−0976)	AI-guided target identification and optimization (Gilead Sciences)	Acetyl-CoA carboxylase inhibition	NASH, metabolic aspects of aging	NCT02876796
GSBR−1290 (aleniglipron)	AI-guided design (Gasherbrum Bio)	GLP−1 receptor activation	Obesity, metabolic syndrome	NCT06703021
OPL−0401	AI-guided design (Valo Health)	ROCK1/2 inhibition	Diabetic retinopathy	NCT05393284

Abbreviations: AI, artificial intelligence; DL, deep learning; DNN, deep neural network; CDK20, cyclin-dependent kinase 20; HCC, hepatocellular carcinoma; HDAC6, histone deacetylase 6; GENTRL, generative tensorial reinforcement learning; DDR1, discoidin domain receptor 1; GANs, generative adversarial networks; mTORC1, mechanistic target of rapamycin complex 1; ML, machine learning; MLP, multilayer perceptron; AI-FM, AI-assisted fluorescence microscopy; GNN, graph neural network; TNIK, TRAF2- and NCK-interacting kinase; A2A, adenosine A2A receptor; RCC, renal cell carcinoma; NSCLC, non-small-cell lung cancer; NASH, non-alcoholic steatohepatitis; GLP-1, glucagon-like peptide-1; ROCK, Rho-associated protein kinase.

conventional screening approaches cannot match.

4. AI-driven innovations in geriatric healthcare

Geriatric healthcare has assumed global urgency as demographic shifts toward an aging population. AI systems, ranging from image recognition, virtual assistants, decision support systems, natural language processing, and emotion detection, underpin many aspects of modern geriatric medicine (Choudhury et al., 2020; Esteva et al., 2019). By integrating continuous streams of physiological, behavioral, and environmental data, these technologies support individualized care and illuminate mechanisms that shape late-life health trajectories. AI addresses core geroscientific challenges, such as monitoring chronic disease progression, detecting early physiological decline, mitigating fall risk, and supporting mental-health resilience (Choudhury et al., 2020). Its integration into clinical workflows advances predictive, preventive, and participatory care models that help extend healthspan and preserve functional independence. Beyond biomedical variables, AI research has elucidated how sociodemographic, socioeconomic, and psychosocial factors, including life satisfaction, social connectivity, environmental quality, and neighborhood infrastructure, modulate aging trajectories (Ruiz-Vanoye et al., 2025). Longitudinal investigations consistently show that these determinants exert greater influence on disease burden and longevity than healthcare expenditure alone. Moreover, AI-driven analyses have identified demographic and socioeconomic predictors of cognitive decline and functional capacity, highlighting the potential of addressing social deprivation to improve nutrition and health outcomes (Ahmadzadeh et al., 2023). Incorporating such variables into predictive models refines our understanding of individual aging processes and enables targeted interventions that improve the quality of life in older adults.

4.1. Smart homes and wearable technology

AI-enabled smart environments and wearable devices have transformed real-time health monitoring in older adults. IoT-integrated environments equipped with adaptive lighting, ambient sensors, and semi-autonomous furniture continuously monitor physiological and behavioral patterns, thereby enhancing independence and safety among older adults (Wang et al., 2025). Complementary wearables such as ECG-smartwatches, biometric textiles, and activity trackers acquire real-time physiological data. These data streams enable early detection of health decline and foster personalized, proactive interventions, directly contributing to improved preventive strategies and advancing our understanding of aging-related health trajectories.

4.2. Predictive analytics and clinical decision support

AI-driven predictive analytics are shifting geriatric care from reactive management to proactive interventions. Supervised and unsupervised ML models analyze complex health data to detect early markers of frailty, fall risk, and disease emergence. These predictive outputs support evidence-based decision-making, guide individualized care strategies and are associated with reduced hospitalization rates (Choudhury et al., 2020). AI also strengthens disease risk stratification through continual model refinement from extensive cohort data and dynamic model updates, aligning with principles of precision medicine. By interacting with genomic and longitudinal health histories, AI systems advance personalized interventions, thereby optimizing therapeutic strategies and anticipating adverse drug interactions rooted in polypharmacy, a pervasive challenge in aging populations (Esteva et al., 2019).

4.3. Robotics in elderly support

Robotics is reshaping caregiving by augmenting mobility, rehabilitation, and emotional support. Automated medication dispensers, mobility-aid robots, and fall-detection systems use multimodal sensors and computer vision algorithms to reduce injury risk and promote independence (Padhan et al., 2023). Rehabilitation robots assist gait training and motor recovery, while humanoid and social robots provide companionship, facilitate daily routines and support communication between older adults and healthcare providers (Nichol et al., 2024). These technological systems support mobility, rehabilitation, and emotional well-being, thereby reducing caregiver workload and improving both physical and cognitive outcomes for older adults. Within caregiving settings, optimized scheduling and interaction-management algorithms ensure timely, personalized assistance. Collectively, these systems complement human caregivers and foster independence and well-being among the aging population.

4.4. Cognitive engagement and mental health support

AI-enabled gamification and virtual-reality (VR) platforms are emerging as potent tools for strengthening neurocognitive resilience and enhancing physical engagement in older adults. Tailored gaming applications and immersive VR environments are now incorporated into post-stroke rehabilitation, dementia care and social-participation programs, where they support motor recovery, stimulate executive and memory functions and reduce behavioral and psychological symptoms (Yang et al., 2025). These systems leverage adaptive difficulty, real-time feedback and embodied interaction to promote neuroplasticity and sustain motivation, key determinants of functional recovery in aging populations. AI also plays an expanding role in mental health support through conversational agents and digital companions. By analyzing speech patterns, interaction dynamics, and affective cues, these systems help detect early cognitive impairment, including prodromal AD, while simultaneously mitigating loneliness and monitoring emotional well-being (Li et al., 2023). Their continuous, context-aware engagement enables more responsive and personalized care, complementing clinical oversight and extending into daily life. Together, AI-driven gamification, VR platforms and affective computing technologies form an integrated framework for cognitive, emotional and social support in late life, advancing geroscience goals by reinforcing resilience and preserving functional independence.

5. Validation imperatives for AI-driven geroscience

AI's translational potential in geroscience depends on systematic validation across three interdependent domains, including data quality, mechanistic interpretability, and reproducibility. Together, these imperatives determine whether computational predictions yield actionable therapeutics or non-generalizable artifacts.

5.1. Data quality

Data quality determines whether AI models extract biological aging mechanisms or amplify technical artifacts, rendering rigorous curation and preprocessing essential prerequisites for translational geroscience (Singh and Benayoun, 2023). Aging omics datasets are uniquely vulnerable because they must capture subtle, progressive molecular changes unfolding over decades while distinguishing them from variation introduced by sex, ancestry, tissue composition, comorbidity, sequencing platform and normalization strategy (Singh and Benayoun, 2023). Longitudinal sampling over months to years further makes contemporaneous processing impossible, introducing batch effects from library preparation and sequencing (Singh and Benayoun, 2023). High-capacity models then encode these confounders as aging patterns, yielding clocks and classifiers that perform well in their training cohorts but fail when externally deployed. Models trained on confounded, batch-affected, or demographically narrow datasets generate outputs that reflect artifact rather than biology, rendering downstream target

nominations and therapeutic predictions unreliable regardless of algorithmic sophistication.

These quality failures manifest across AI geroscience applications. Meta-analysis of RNA-seq–based aging and pro-longevity signatures shows that inter-laboratory variance can dominate signal, undermining model generalizability and misdirecting target discovery when models trained in single cohorts are applied to new populations (Bairakdar et al., 2023). Similarly, methylation-based aging clocks demonstrate variable performance across tissues, ancestries, and disease states, raising questions about whether they capture conserved aging mechanisms or cohort-specific epigenetic drift (Kriukov et al., 2025; Richardson et al., 2025). When such confounded models are used to select hallmark pathways or senolytic-responsive subgroups, errors propagate through the drug development pipeline and bias therapeutic investment toward non-generalizable biology.

Minimizing these data quality failures requires interventions at the experimental design stage rather than relying on post-hoc correction. Study designs that randomize biological groups across batches and platforms reduce confounding at the source, while downstream batch-correction must be paired with sensitivity analyses to verify that aging signals survive adjustment (Singh and Benayoun, 2023). Adversarial training that explicitly models and regresses out batch factors, transfer and domain-adaptation methods that align feature distributions across cohorts, and federated learning frameworks that train shared models across institutions without data centralization help preserve biological signal while leveraging diverse populations (Danek et al., 2024; Singh and Benayoun, 2023). Generative AI can further augment underrepresented demographic or biological strata with synthetic data, alleviating sparsity that would otherwise cause aging models to conflate cohort structure with age-related biology (Singh and Benayoun, 2023).

5.2. Mechanistic interpretability through explainable AI

AI models that predict biological age or geroprotective targets without mechanistic grounding cannot distinguish causal aging drivers from statistically correlated bystanders, a distinction that determines whether predictions translate into functional therapeutics or spurious associations (Fuellen et al., 2025; Kriukov et al., 2025). This gap has fueled the proliferation of aging clocks that achieve impressive cross-validation metrics but yield discordant age estimates and fail to predict intervention responses, undermining confidence in AI-guided geroscience (Kriukov et al., 2025).

XAI methods address this limitation by functioning as pre-experimental filters that anchor model predictions to known aging hallmarks. Techniques such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) attribute predictions to specific features, with SHAP providing cohort- and individual-level importance scores grounded in cooperative game theory (Gašperlin Stepančič et al., 2024; Hancock et al., 2025; Ponce-Bobadilla et al., 2024; Salih et al., 2025). LIME offers localized surrogate models that clarify specific predictions, helping clinicians understand how biomarkers, functional tests, or lifestyle factors drive risk estimates (Hettikankanamage et al., 2025). In proteomic aging clocks, SHAP analysis shows that ECM remodeling proteins, inflammatory cytokines, and metabolic enzymes dominate predictions, mechanistically embedding these signatures within established geroscience frameworks and prioritizing pathways for experimental validation (Argentieri et al., 2024; Kuo et al., 2024). By clarifying which molecular features drive predictions, XAI helps separate candidate therapeutic targets from merely correlative biomarkers, while also informing feature selection, reducing overfitting, and improving reproducibility across aging populations (Qadri et al., 2025). These tools have demonstrated value in modeling conditions such as sarcopenia and cognitive decline, where pathway-level insight guides targeted interventions (Kim et al., 2021; Loo et al., 2025).

5.3. Hierarchical experimental validation

Mechanistic interpretability through XAI must be coupled with rigorous experimental validation to ensure computational predictions translate into functional therapeutics. Recent AI-target frameworks treat validation as assembling an auditable evidence chain, advancing candidates only when convergent data from human genetics, disease-context specificity, tractability, on-target safety, biomarkers, and perturbation–rescue studies support modulation of that node (Fuellen et al., 2025; Pun et al., 2026). In geroscience, this chain must map onto a hierarchy that moves predictions from in silico pathway analysis through senescence, autophagy, or mitochondrial assays into lifespan and healthspan readouts in model organisms before clinical deployment, because aging is a system-level process whose modulation demands benefit across biological scales. As highlighted in previous sections, several AI-generated hypotheses have already traversed this pipeline from prediction to invertebrate and mammalian validation and into early-phase human trials, underscoring that coupling XAI-guided mechanistic insight to a geroscience-specific validation hierarchy is key for turning raw model outputs into therapeutically viable interventions.

5.4. Reproducibility and transparent reporting

Reproducibility failures disproportionately undermine geroscience AI because models that appear robust within a single cohort can give contradictory answers when applied elsewhere, making it unclear whether they capture conserved aging mechanisms or local noise (Singh and Benayoun, 2023). A recent appraisal found that about 38% of AI-based aging clocks lack proper validation and yield discordant biological age estimates across methods, limiting their usefulness for trial enrichment and intervention monitoring((Zhang et al., 2026), (Kriukov et al., 2025)). Moreover, chronological age prediction accuracy does not correlate with mortality prediction, exposing that current benchmarking metrics fail to capture the biological relevance geroscience applications demand (Ying et al., 2025). Broader surveys of AI projects reinforce this concern, with data and process quality identified as primary failure drivers, suggesting that without stronger safeguards, AI may entrench rather than reduce clinical attrition (Singh, 2026).

Addressing these failures requires explicit reproducibility safeguards layered on top of data-quality and interpretability standards. Transparent reporting of neural architectures, hyperparameter search strategies, data preprocessing pipelines, feature engineering steps, and negative results forms the foundation. AI-specific reporting standards extending TRIPOD + AI (Collins et al., 2024) and CONSORT-AI (Liu et al., 2020) guidelines enable independent verification that models capture conserved aging mechanisms rather than cohort-specific correlations. Recent FDA guidance on AI-enabled software further treats context-of-use definitions, performance monitoring, drift detection, and retraining plans as regulatory expectations, making reproducibility a compliance requirement as well as a scientific one (FDA, 2025). Preregistration of analysis plans and public deposition of code and models in repositories such as Hugging Face or ModelHub can allow independent groups to stress test geroscience models, converting AI outputs into a cumulative and verifiable body of aging biology rather than a catalog of non-reproducible predictions.

6. Ethical, social and infrastructure considerations

Beyond the technical validation, translating AI-geroscience into equitable practice requires confronting ethical, social, and infrastructural barriers that disproportionately affect vulnerable older adults. Although AI can accelerate target discovery, refine aging trajectories, and personalize interventions, its deployment in geroscience raises concerns about transparency, fairness, privacy, and autonomy that are especially acute for underrepresented or marginalized populations. The

transformative impact of AI in aging research depends on addressing model robustness, algorithmic bias, infrastructure limitations and societal implications.

6.1. Algorithmic bias

AI systems trained on historical datasets often encode implicit human biases, reinforcing societal inequalities in geroscience models (Cajueiro and Celestino, 2026; Chu et al., 2022). These biases shape exposure profiles, healthcare access, and environmental stressors across socioeconomic, racial, ethnic, gender, and disability groups, which modulate aging trajectories and biological risk. Biological age estimators and risk stratification tools can therefore misclassify marginalized individuals, particularly in settings of accelerated aging or limited preventive care (Chu et al., 2022; Panch et al., 2019). Model accuracy further deteriorates when training data lacks phenotypic, behavioral, and environmental heterogeneity. Representation bias, well documented in face recognition deficits for women and people of African descent, has clear analogs in aging clocks and multimodal risk models, which, when trained on imbalanced datasets, generate biased estimates of biological age and disease susceptibility in under-sampled or structurally disadvantaged groups (Cajueiro and Celestino, 2026; Ganel et al., 2022; Klare et al., 2012). Mitigation requires interventions across the pipeline, from rebalancing datasets and adding fairness constraints during training to cautious use of synthetic augmentation and age- and ancestry-specific validation, so that mechanistically grounded AI tools reflect genuine biological heterogeneity rather than entrenching existing disparities (Chu et al., 2023).

6.2. Digital agism

Agism encompasses prejudice, discrimination, and systemic exclusion rooted in age-based assumptions (Chu et al., 2023; Nyrup et al., 2023). When combined with resource inequalities and algorithmic biases, these forces catalyze digital ageism, where technological systems amplify age-related disparities (Chu et al., 2023). Digital ageism arises not only from biased datasets and crude proxy variables, but also from contextual factors such as developer demographics, data provenance, and interpretive frameworks (Chu et al., 2022; Chu et al., 2023; Nyrup et al., 2023). Underrepresentation of older adults in research and design, together with youth-centric assumptions, produces tools poorly matched to age-related physiology, cognition, and mobility, which impairs usability, distorts aging data, and reinforces ageist stereotypes (Chu et al., 2023; Nyrup et al., 2023). These dynamics threaten biological age estimates, functional assessments, and clinical decisions in geroscience and shape social determinants of health, including employment, healthcare, and social participation, thereby reinforcing narratives of diminished agency in later life. Addressing digital ageism therefore requires broader demographic representation in datasets, routine bias detection, critical review of embedded assumptions, and more inclusive digital environments, so that AI systems do not compound intersecting oppressions such as sexism, racism, and classism across aging populations (Chu et al., 2023; Nyrup et al., 2023).

6.3. Clinical trust, accountability and regulatory oversight

As global aging accelerates and geroscience integrates AI into clinical workflows, trust and accountability become central, especially when model outputs influence medication choices, fall-risk assessments, and screening protocols in older adults (Haibe-Kains et al., 2020). Although DNNs and transformer models often outperform traditional algorithms in prediction accuracy, their opaque functioning can undermine clinical trust and lead to diagnostic errors, particularly among individuals with complex aging trajectories (Hassija et al., 2024; Kalyakulina et al., 2024). Lack of transparency risks perpetuating bias and diagnostic errors, undermining care among individuals with complex aging trajectories. XAI techniques offer needed transparency, but interpretability alone is insufficient without robust governance. Layered explainers can themselves add complexity, which reinforces the need for human oversight to check outputs and preserve accountability. Clinicians must understand not only which features drive a prediction but also whether that output is biologically plausible and consistent with clinical judgment. As regulatory frameworks evolve, explicitly addressing liability, transparency, and mechanistic accuracy is essential. The FDA's 2025 guidance on AI-enabled device software functions makes context-of-use documentation and lifecycle maintenance regulatory obligations, requiring ongoing monitoring and updating of deployed models (FDA, 2025). This extends beyond reproducibility to clinical safety, ensuring that AI systems used in geroprotector development or geriatric care retain performance as populations, practice patterns, and biological knowledge evolve. International bodies such as the WHO and privacy frameworks like the General Data Protection Regulation further reinforce ethical, transparent governance that centers patient autonomy and informed consent (Kashefi et al., 2024). Ultimately, integrating technical interpretability with regulatory oversight and clinical accountability is essential for maintaining trust and upholding ethical standards in AI-driven geroscience.

6.4. Privacy and accountability

Geroscience increasingly depends on longitudinal, multi-institutional datasets that link genomic, behavioral, and clinical data, making privacy and informed consent especially critical for older adults whose lives are heavily medicalized and datafied. Digital twins, high-resolution computational models of biological systems, raise concerns about consent, data sovereignty and third-party access to simulations encoding sensitive biological features (Huang et al., 2022). Surveillance-oriented AI further complicates privacy by extracting granular behavioral and biometric signals, threatening autonomy and enabling intrusive profiling (Rathore, 2026). Even in pharmacogenetics, where warfarin dosing algorithms for older atrial fibrillation patients combine genotypic, demographic, and clinical variables to reduce bleeding and thrombotic risk, supposedly de-identified datasets remain vulnerable to re-identification using auxiliary information from public records or social media (Dietz et al., 2020; Ren et al., 2020).

The core challenge is to extract population-level insight without exposing individuals amid pervasive auxiliary data sources. Differential privacy can protect identities by adding calibrated noise while preserving analytical utility, especially when standard anonymization fails (Wang et al., 2023). Federated learning and cryptographic approaches reduce exposure by supporting decentralized training, while privacy-by-design, data minimization, re-engineered consent, and privacy impact assessments embed protection into AI systems (Cajueiro and Celestino, 2026; Taherdoost et al., 2025). Governance mechanisms such as data trusts can further clarify stewardship and access rights. Balancing these safeguards with computational efficiency is essential: excessive noise can obscure key aging biomarkers, whereas weak protection risks discrimination against vulnerable groups (Cajueiro and Celestino, 2026). Ultimately, integrating stratified data classification, targeted privacy protections, and lifecycle-aware governance is essential to maintaining trust, reducing disparities, and upholding ethical standards in geroscience.

6.5. Infrastructure

Quantum computing and autonomous AI could transform aging research by enabling high-throughput molecular modeling and hybrid classical–quantum workflows that accelerate the identification of viable geroscience targets (Pineda et al., 2026). Autonomous systems that generate hypotheses, design experiments, and draft manuscripts are already entering industrial pipelines (Shen et al., 2025; Wang et al., 2023). Nonetheless, large-scale models, such as transformer-based drug

discovery engines, diffusion models, and multimodal predictors, necessitate computational resources, energy demands, GPU availability, and technical expertise that many aging research centers lack (Sundberg and Holmström, 2023). Scalable solutions, including lightweight architectures, distributed computing frameworks, and open-source platforms, are vital to democratize access.

Deployment remains challenging within fragmented clinical IT ecosystems. Inconsistent EHRs, heterogeneous biomonitoring data streams and incompatible data standards impede integration. Without standardized data pipelines for data ingestion, preprocessing and harmonization, real-time model updating and drift detection are impractical, impeding clinical translation. Moreover, multidisciplinary collaboration gaps compound these issues, where data scientists optimize architectures, while geroscientists confront multimorbidity, polypharmacy and cohort-specific physiological heterogeneity. When shared infrastructure or comprehensive datasets are lacking, these groups struggle to converge on models that capture the mechanistic complexity of aging, further hindering progress (Sundberg and Holmström, 2023).

Low- and no-code AI platforms like BigML, Google AutoML, and AzureML reduce technical barriers but still rely on secure cloud infrastructure, interoperable data pipelines, and rigorous validation, resources often absent in geriatric care settings (Sundberg and Holmström, 2023). Progress hinges on standardized repositories such as Geroprotectors.org (Moskalev et al., 2015) and DrugAge (Belikov et al., 2025), which curate lifespan-extending interventions and elucidate underlying molecular mechanisms. These resources are indispensable for training, benchmarking, and validation in geroscience. Ultimately, without resilient, interoperable, and well-maintained infrastructure, even the most sophisticated algorithms cannot reliably inform aging interventions or ensure equitable clinical translation.

7. Conclusion

The convergence of geroscience and AI is reshaping how aging is conceptualized, measured, and targeted. By integrating multi-omics, imaging, clinical, and behavioral data across biological hierarchies, AI clarifies how primary, antagonistic, and integrative hallmarks interact to produce tissue-specific vulnerability and systemic decline. This systems perspective recasts aging as a dynamic, interconnected process rather than a collection of isolated traits. Along the discovery pipeline, AI accelerates target identification and mechanistic dissection, reveals regulatory nodes and intercellular communication networks, and proposes multi-target therapeutics and synergistic combinations that shorten the route from target nomination to in vivo validation. Rational repurposing is further enabled by detecting molecular convergence across age-related disorders, while at the clinical edge, predictive analytics refine risk stratification for frailty, multimorbidity, and cognitive decline. Collectively, these capabilities represent not an incremental refinement of existing tools but a qualitative expansion of what geroscience can ask, resolve, and act upon.

However, the durability of these advances remains contingent on resolving fundamental scientific, ethical, and infrastructural challenges that will ultimately determine whether AI fulfills or forfeits its promise in geroscience. Data quality failures, limited mechanistic interpretability, and poor reproducibility can convert powerful models into non-generalizable artifacts, and without rigorous curation and validation hierarchies, AI risks encoding confounders as biological aging signals rather than revealing them. Algorithmic bias, digital ageism, and youth-centric design threaten to widen disparities for populations already experiencing accelerated aging, while continuous biomonitoring intensifies concerns around privacy, consent, and data sovereignty. Validated candidates must therefore pass through geroscience-specific hierarchies spanning in silico pathway analysis, senescence and autophagy assays, organismal lifespan and healthspan studies, and only then early-phase clinical trials. Next-generation biological age metrics need causal grounding in perturbation screens, senolytic responses, and metabolic flux, and federated learning on diverse global datasets must replace narrow cohort training as the field's standard. An AI-enabled geroscience ecosystem must be architected for equity and trust from the outset, through transparent reporting, lifecycle governance, and infrastructure that smaller centers and low-resource settings can realistically operate.

Geroscience has long sought to treat aging itself rather than its downstream consequences, one disease at a time. AI now makes that ambition technically tractable, offering tools to map the causal architecture of aging, identify intervention points shared across hallmarks, and match individuals to therapies with a precision that population-level medicine cannot achieve. Whether the convergence of AI and geroscience becomes a genuine turning point depends not on algorithmic power alone but on the rigor with which models are built, the breadth of the populations they represent, and the honesty with which their limitations are disclosed. By meeting these standards, AI-enabled geroscience will not merely illuminate the biology of aging but will stand, for the first time in human history, at the threshold of transforming aging from an inevitability into a modifiable biological process.

Author contribution

H.L., Q.Y. and L.W.N. conceived and designed the study. H.L. and P. O.A. conducted the initial literature search and wrote the original draft. Q.Y., W.C., and L.W.N. revised and finalized the manuscript. All authors contributed to the article and approved the submitted version.

Declaration of Generative AI and AI-assisted technologies in the writing process

While preparing this manuscript, the authors utilized Microsoft Copilot to enhance language clarity, conciseness, readability, and stylistic consistency. The content was thoroughly reviewed, revised, and validated by the authors, who bear full responsibility for the accuracy and integrity of the final version.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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